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Preliminary Mapping of Deep-Seated Landslides in the Washington Coast Range, Cascadia Subduction Zone

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Cascadia CoPes Hub — The Cascadia Coastlines and Peoples Hazards Research Hub
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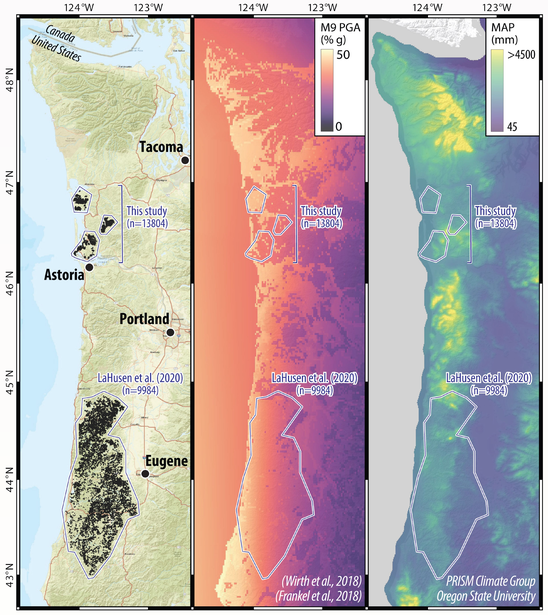
Main Objectives

- Fill the landslide inventory gap in the Washington Coast Range.
- Investigating the effectiveness of various surface complexity measures in relative dating on the deep-seated landslide events.
- Understanding the spatio-temporal patterns of landslides along the Cascadia Subduction Zone (CSZ), by comparing with Oregon dataset.
- Studying the primary controls on landsliding, possible trigger(s), the style of failure, the characteristics of its deposits, and their long-term impact(s) to the landscape evolution and sediment routing.

Current Findings

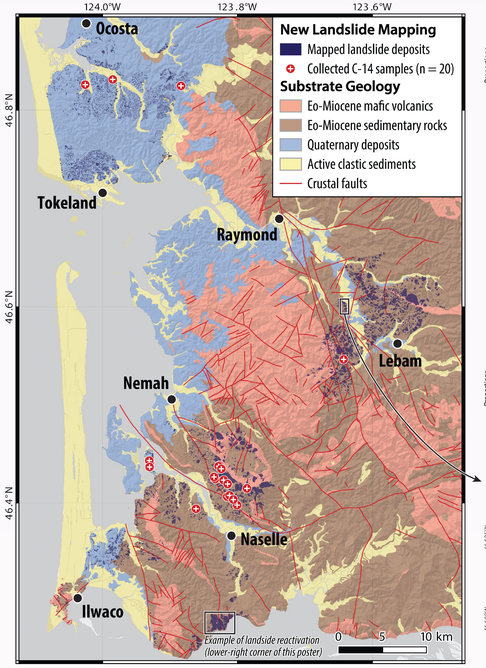
- Beckrock lithology plays a primary role in controlling the size, style, and density of landsliding.
- No statistically significant relationship has been identified between landslide area density *vs.* simulated CSZ M9 Earthquake peak ground accelration (PGA) or 30-year mean annual precipitation (MAP).
- It remains mysterious about why there is a considerable difference in the landslide size & density within a similar lithology (Eo-Miocene marine sedimentary rocks) across Orogen and Washington. Detailed data classification of the landslides and/or consistency of the mapping criteria needed to be re-examinated.

Cascadia WA-OR Coastal Mountains



WA-OR landslide inventory produced by our team to date, plotted against M9 CSZ earthquake PGA and mean annual precipitation (MAP).

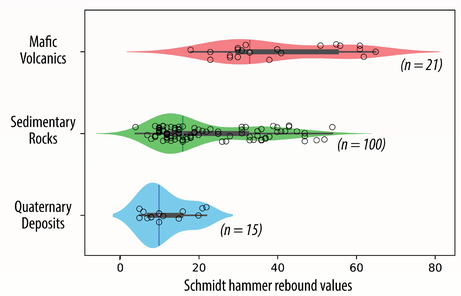
Study Area in SW Washington



Simplified geological map of the southwestern Washington, overlaid with the landslide mapping progress to date and locations of collected C-14 dating materials from mapped landslide deposits.

Bedrock Strength

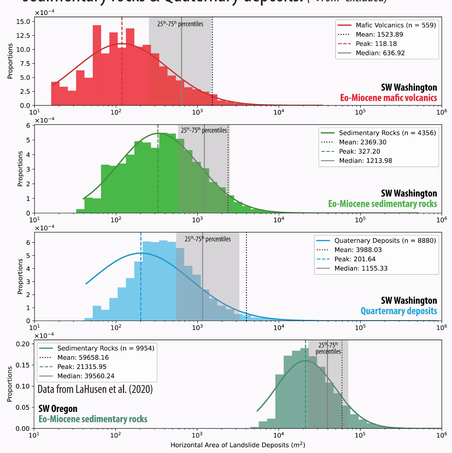
Schmidt hammer measurements *(WA data only)*



Distinguishable mechnical properties among three major rocktypes in southwestern Washington.

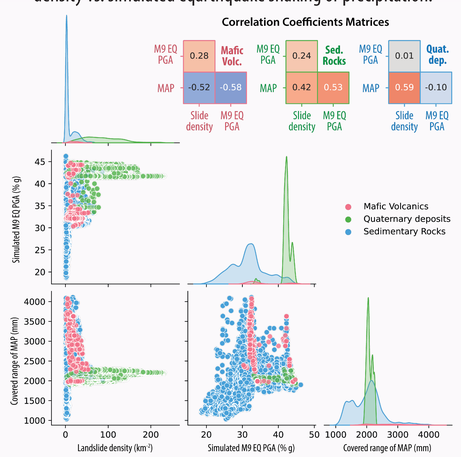
Landslide size *vs.* substrate lithology *(WA + OR data)*

In WA, landslides in volc. rocks are generally smaller than sedimentary rocks & Quaternary deposits. *(<10m² excluded)*

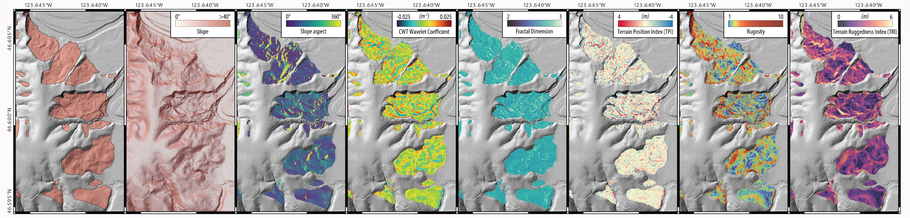


Landslide density *vs.* M9 PGA *vs.* MAP *(WA + OR data)*

No statistically significant relationship between landslide density vs. simulated eqarthquake shaking or precipitation.

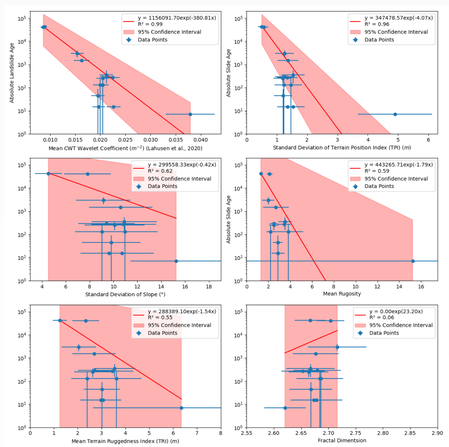


Testing various measures of surface complexity *(Example near Lebam, WA)*



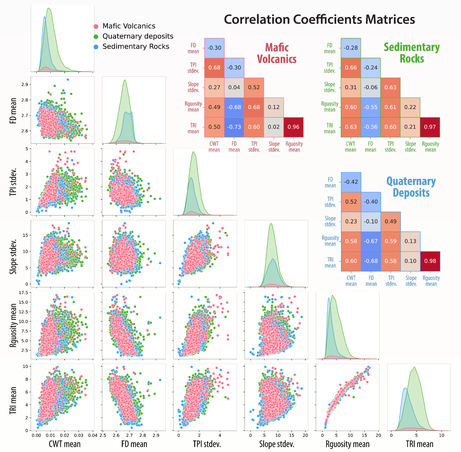
Surface complexity measures *vs.* dated absolute ages

(OR data only) CWT Wavelet Coefficient remains the most suitable measure for landslide topographic relative dating!



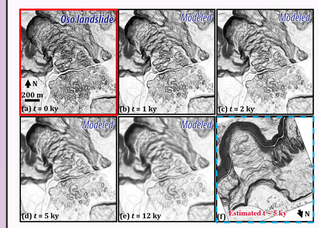
Cross-correlation among surface complexity measures

(WA + OR data) Weak positive correlation among different surface complexity measures (except *Fractal Dimension*).



(NEXT STEP!) Surface complexity based relative dating

Previous workers have successfully applied the surface complexity measure of larger-sized, deep-seated landslide deposits as a proxy for relative dating of landsliding events in other places. This method is based on an assumption that the surface of a landslide deposit will smooth out profressively by diffusive surface processes.



(Booth et al., 2017)

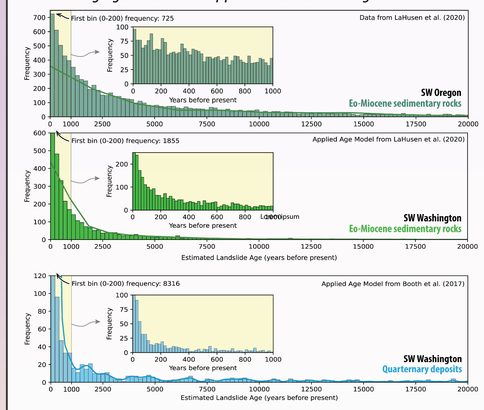
Successful example from North Fork (Booth et al., 2017)Stillaguamish River valley (NFSR), WA

(a-e) Slope maps of the surface evolution of the 2014 Oso landslide predicted by a nonlinear hillslope diffusion model.

(f) Estimated depositional age of a similar-sized ancient landslide nearby, based on surface complexity measurement using LiDAR digital elevation model data.

Landslide frequency through time...

**Assuming age models are applicable across the region...*



Other concerns to be investigated...

- Commonly observed features of landslide reactivation and earth-flow style landslides may create bias in geomorphic mapping and uncertainties in surface-complextiy relative dating, particularly in regions with sedimentary rocks.
- Potential inconsistent mapping criteria.

Acknowledgement & Reference list

